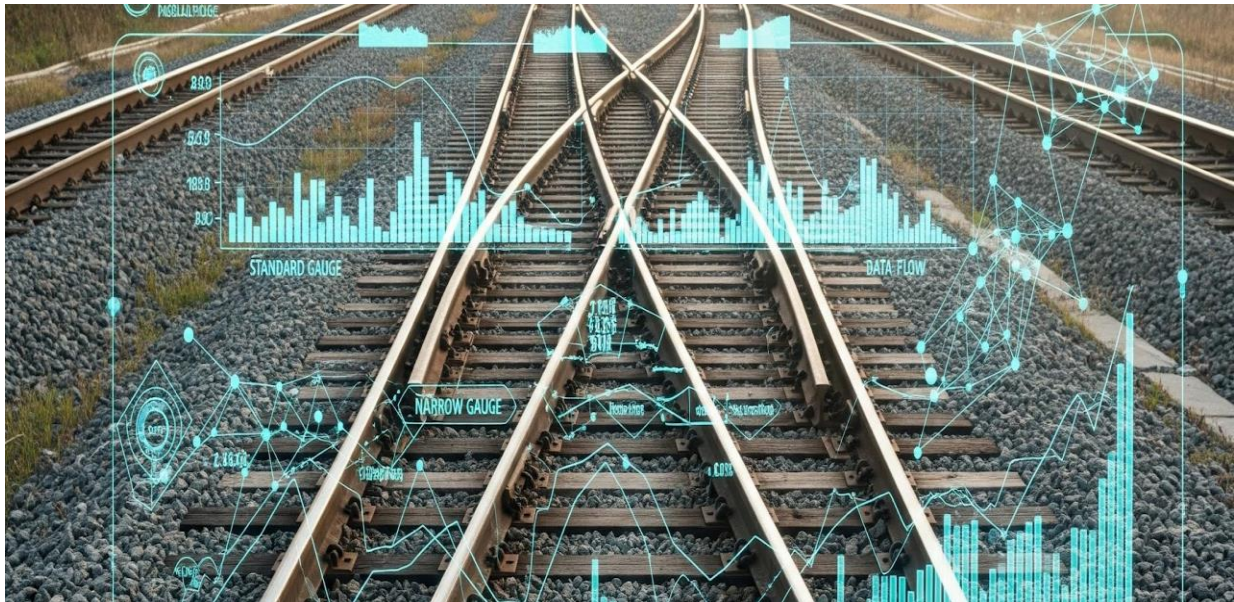




Data Analytics Project: Railway Gauge Trends

Dataset Used: _____railway_gauges.csv_____

This document outlines a structured, end-to-end data analysis project based on railway gauge data. The project is divided into two main sections: **Data Analytics Project: Railway Gauge Trends**

This document outlines a structured, end-to-end data analysis project based on railway gauge data.

Dataset: Railway Gauge Data (Assumed to be in a CSV file)

Project Goal: To analyze historical trends and composition of different railway gauges (Broad, Metre, Narrow) over the years.**Modules Used**

The **primary libraries/modules** required for this project are:

- **Pandas:** For all data manipulation, filtering, aggregation, and feature engineering tasks.
- **NumPy:** Specifically for numerical calculations like yearly growth (`np.diff`).
- **Visualization Library (e.g., Matplotlib):** For generating all required plots (Line Graphs, Bar Charts, Pie Charts, Stacked Bar Charts).

□ Project Coverage

Category	Skills Covered
Data Preprocessing	Data Loading, Missing Value Handling, Type Conversion
Pandas Operations	Filtering, Selection, Aggregation, Feature Engineering
NumPy Calculations	Yearly growth calculation (np.diff)
Visualization	Line Graph, Bar Chart, Pie Chart, Stacked Bar Chart

Scenario 1: Basic Data Loading & Cleaning (□ Easy)

Description:

The railway gauge dataset contains yearly information about Broad Gauge, Metre Gauge, Narrow Gauge, and Total railway tracks. In this scenario, the dataset is loaded into a Pandas DataFrame and its structure is examined. Missing values are checked and replaced with zero. The relevant gauge columns are then converted into numeric data types for further analysis

Task	Description
1.	Load the dataset into a Pandas DataFrame.
2.	Display the first 5 rows and column names to understand structure.
3.	Check for missing values and replace them with 0 (assuming missing values represent 0 km of track).
4.	Convert all relevant gauge columns (Broad, Metre, Narrow, Total) to numeric types (e.g., float64 or int64).

Graph (Optional):

(Describe the graph and mention the graphs here)

(Insert graph here if applicable else remove the Graph heading and the graph info)

Conclusion:

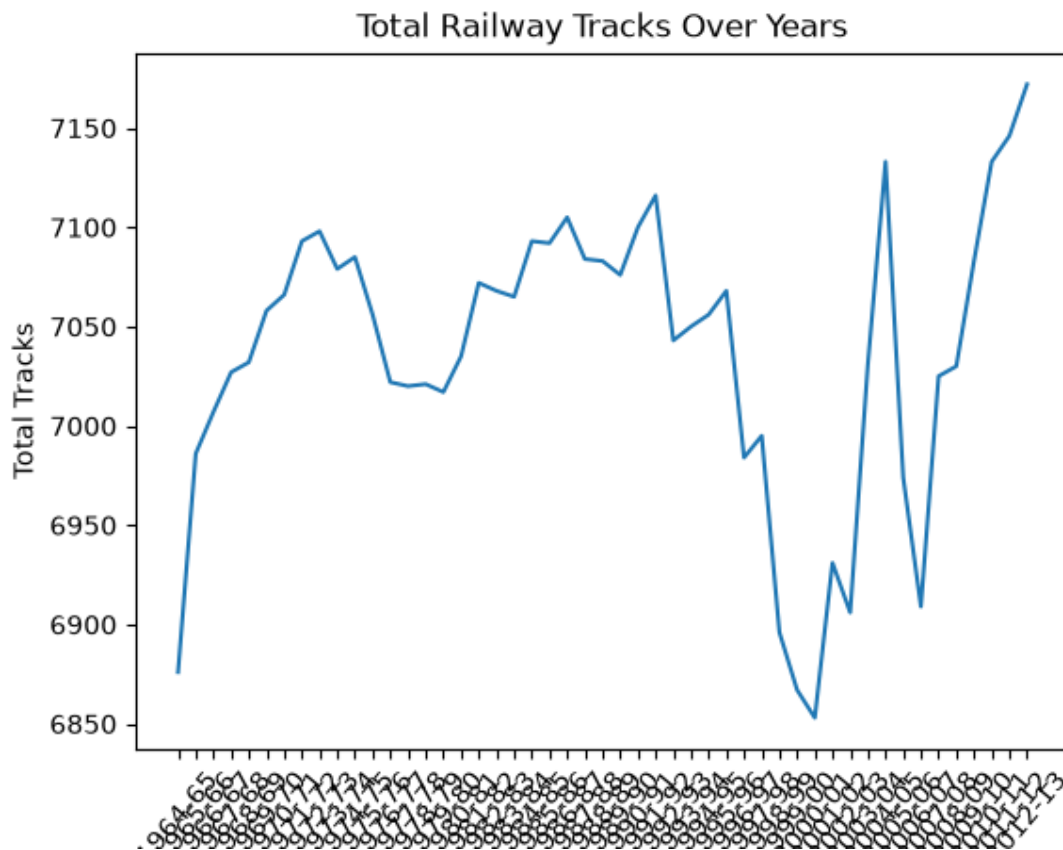
The railway gauge dataset was successfully loaded into a Pandas DataFrame. The first five rows and column names were examined to understand the structure of the data. No missing values were found in the dataset. The Broad Gauge, Metre Gauge, Narrow Gauge, and Total columns were already in numeric form, making the dataset ready for further analysis.

Scenario 2: Simple Visualization (Easy)

Description:

This scenario analyzes the overall trend of the total railway track network over the years. The Year and Total columns are extracted from the dataset for analysis. A line graph is created to visualize how the total railway tracks changed over time. The graph is given a suitable title and axis labels, and the year labels are rotated for better readability. Finally, the first and last Total values are compared to identify whether the overall trend is increasing, decreasing, or unchanged.

Task	Description
1.	Extract the Year and Total columns.
2.	Plot a line graph showing Total tracks over the Years .
3.	Add a descriptive title (e.g., "Growth of Total Railway Network Over Time") and X/Y axis labels.
4.	Identify whether the overall trend is increasing or decreasing and note the finding.



Conclusion:

The line graph shows that the total railway track length has fluctuated over the years, with both increases and decreases during different periods. The total railway tracks were 6876 km in 1964-65 and increased to 7172 km in 2012-13. Therefore, by comparing the first and last values, the overall trend of the total railway network is increasing.

Scenario 3: Filtering + Bar Chart (□ Medium)

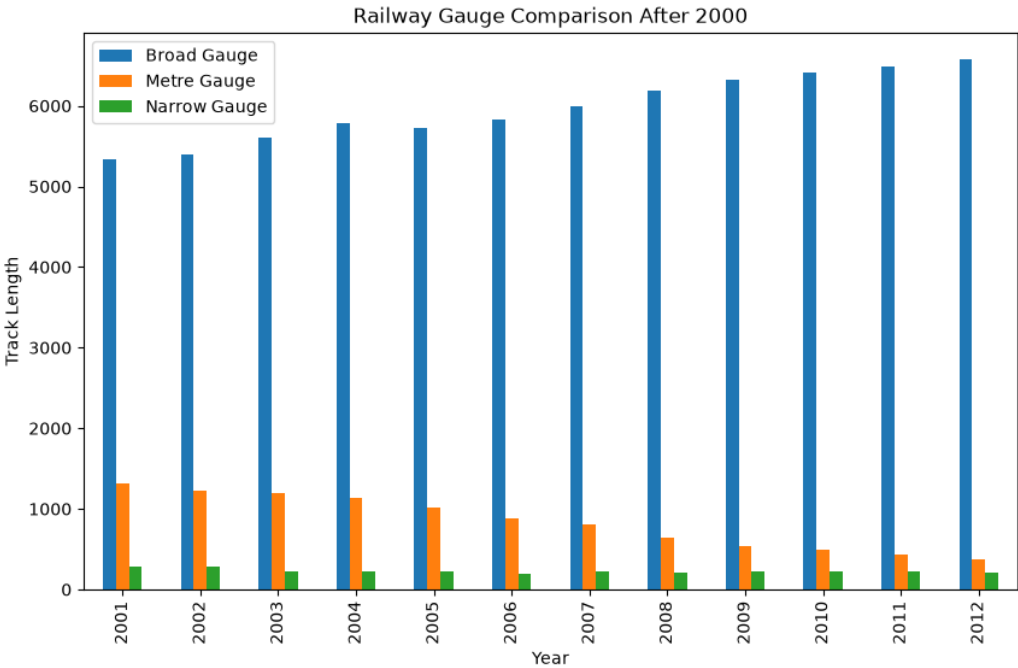
Description:

This scenario analyzes the railway gauge trends for the years after 2000. The starting year is extracted from the Year column and the dataset is filtered accordingly. Broad Gauge, Metre Gauge, and Narrow Gauge are compared using a grouped bar chart. The total track length of each gauge is also calculated to identify the dominant gauge in recent years.

Task	Description

1.	Filter the dataset to include only years <i>after</i> 2000.
2.	Select the Broad Gauge, Metre Gauge, and Narrow Gauge columns for the filtered data.
3.	Plot a grouped bar chart comparing the three gauges for the selected years.
4.	Add a legend and ensure proper labels for axes.
5.	Identify which gauge dominates in recent years (post-2000).

Graph (Optional):



Conclusion:

The analysis shows that Broad Gauge has the highest total track length after 2000, while Metre Gauge and Narrow Gauge have smaller contributions. Among the three gauge types, Broad Gauge clearly dominates the railway network in recent years. The grouped bar chart also shows the significant difference between Broad Gauge and the other two gauges. Therefore, Broad Gauge is the dominant gauge in the post-2000 period.

Scenario 4: Feature Engineering + Pie Chart (□ Medium)

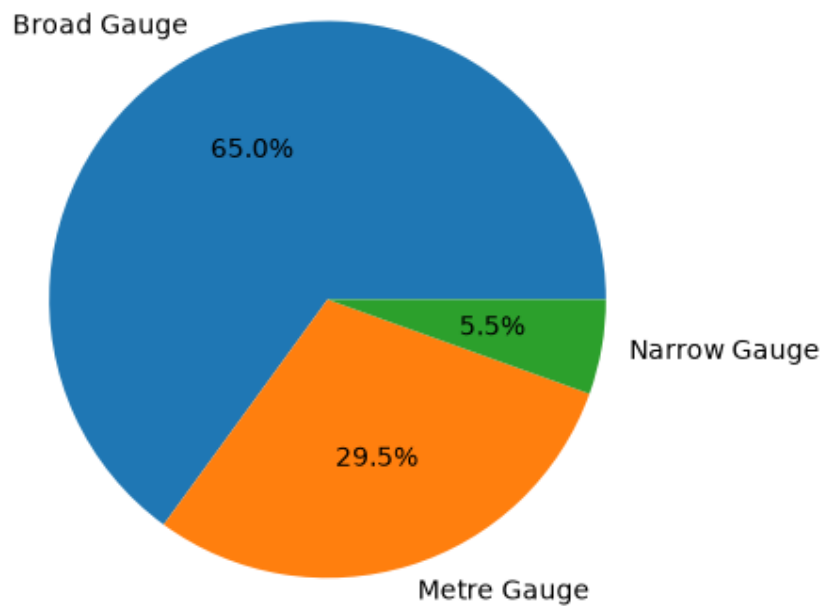
Description:

This scenario analyzes the overall contribution of each railway gauge throughout all available years. The total track length of Broad Gauge, Metre Gauge, and Narrow Gauge is calculated and stored in a Pandas Series. A pie chart is then used to visualize the percentage contribution of each gauge type. The gauge with the highest total contribution is identified to determine the most dominant gauge in the railway network.

Task	Description
1.	Calculate the total sum of each gauge (Broad Gauge, Metre Gauge, Narrow Gauge) across all available years.
2.	Create a new structure (Pandas Series or single-column DataFrame) holding these three total values.
3.	Plot a pie chart using this new structure to show the percentage contribution of each gauge type.
4.	Add percentage labels to the slices using autopct (e.g., '%.1f%%').
5.	Interpret which gauge contributes the most to the entire history of the railway network.

Graph (Optional):

Contribution of Each Railway Gauge



Conclusion:

The pie chart shows that Broad Gauge contributes the largest share of the railway network across the entire historical period. Metre Gauge has the second-highest contribution, while Narrow Gauge contributes the least. Therefore, Broad Gauge is the gauge with the highest overall contribution to the railway network.

Scenario 5: Advanced Analysis + Multiple Graphs (□ Hard)

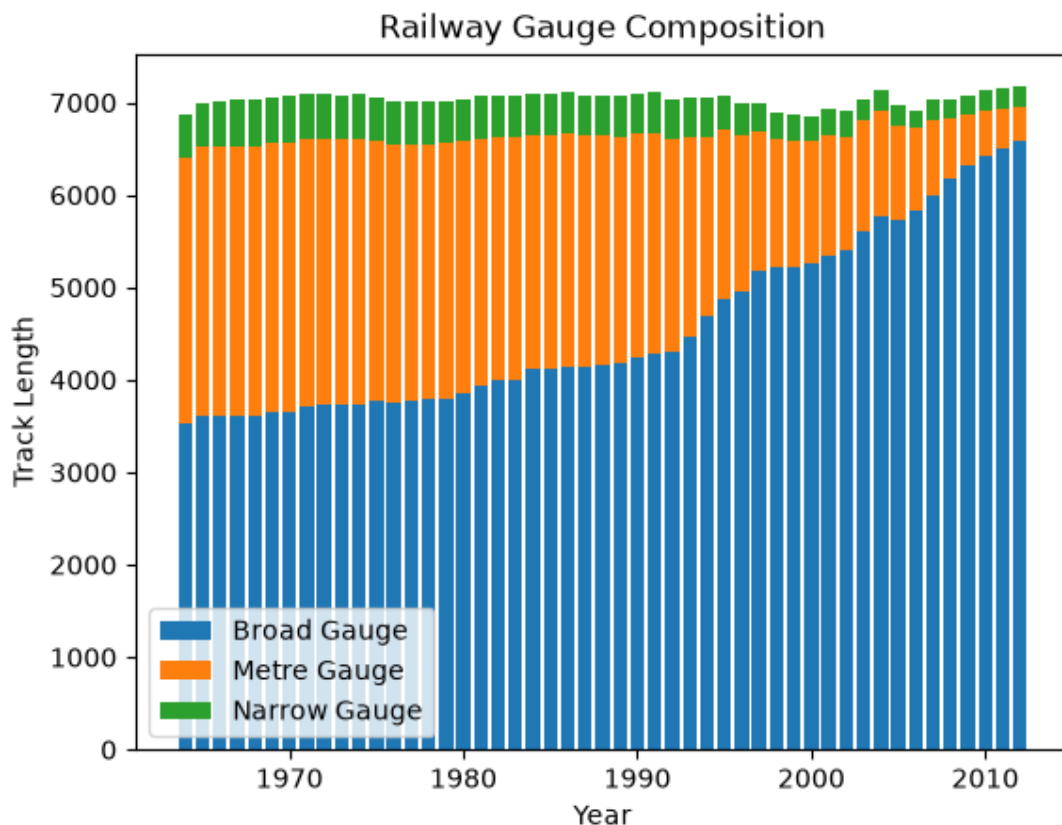
Description:

This scenario performs an advanced analysis of railway gauge trends over the entire period. Percentage composition columns are created for Broad Gauge, Metre Gauge, and Narrow Gauge relative to the Total tracks, while NumPy `np.diff()` is used to calculate yearly changes in Total tracks. A line graph and stacked bar chart are created to understand gauge trends and their composition over time. The analysis also checks for declines in individual gauges and examines whether the railway system is shifting towards a single dominant gauge.

Task	Description
1.	Create new columns for percentage composition relative to Total tracks for each year: % Broad Gauge, % Metre Gauge, and % Narrow Gauge.

2.	Use the NumPy function np.diff to calculate the yearly growth (change) of the Total tracks column.
3.	Plot: a) A single Line Graph showing the trend of all three percentage composition columns (% Broad Gauge, % Metre Gauge, % Narrow Gauge) over the years. b) A Stacked Bar Chart showing the raw composition of the three gauges (Broad, Metre, Narrow) over time.
4.	Highlight/Identify: a) The year(s) with the highest yearly growth in Total tracks. b) Any specific decade or period showing a significant decline in any particular gauge.
5.	Final Conclusion: Based on the percentage composition line graph, provide a conclusion to the question: "Is the railway system shifting towards a single dominant gauge?"

Graph (Optional):



Conclusion:

The analysis shows that the railway gauge network has experienced both growth and decline during different periods. Broad Gauge remains the most dominant gauge, while Metre Gauge and Narrow Gauge have smaller contributions. The percentage

composition indicates a strong presence of Broad Gauge throughout the period. Overall, the railway system shows a shift towards Broad Gauge as the single dominant gauge

Final Conclusion:

This project analyze railway gauge trends from 1964-65 to 2012-13 using Pandas, NumPy, and Matplotlib. The analysis showed fluctuations in the total railway network, with an overall increasing trend from the beginning to the end of the period. Broad Gauge consistently had the highest contribution compared with Metre and Narrow Gauge. The visualizations also showed that all three gauges experienced some periods of decline. Overall, the railway system shows a strong shift towards Broad Gauge as the dominant gauge.

What you have learned

Through this project, I learned how to load, clean, and analyze a real-world railway gauge dataset using Pandas. I learned how to handle missing values, filter and select data, calculate totals and percentages, and use NumPy `np.diff()` to find yearly growth. I also learned how to create and interpret different visualizations such as line graphs, grouped bar charts, pie charts, and stacked bar charts using Matplotlib. Finally, I learned how to identify trends, compare railway gauges, find declines and dominant gauges, and draw meaningful conclusions from data.